



AutoJudge

Frontend page specification — Angular

CONTENTS

- 01 Dashboard + system control
- 02 Assignments list
- 03 Grade assignment
- 04 Results view (overall)
- 05 Rejudge
- 06 Test case manager
- 07 Config editor
- 08 Plagiarism results

01 · overview

Dashboard

Workers online

4

Result worker

● Running

Queued jobs

17

Last batch

2 min ago

Worker pool

pids-limit 100 · timeout 2s



4

Add worker

Remove worker

Start everything ↗

Active workers

worker-01	grading DSA-A3, student 21i-0214	●
worker-02	idle	●
worker-03	grading DSA-A3, student 21i-0219	●
worker-04	idle	●



02 · GET /api/assignments

Assignments

Assignment	Course	Language	Submissions	Status
Assignment 3 — Linked Lists	DSA	cpp	45	GRADED
Assignment 4 — Graphs	DSA	cpp	45	GRADING...
Assignment 2 — Recursion	OOP	java	38	GRADED
Assignment 1 — Pointers	PF	cpp	61	NOT GRADED

Source: scanned assignment directories · each row links to Grade / Results / Rejudge / Config for that assignment

03 · POST /api/assignments/{id}/grade

Grade — Assignment 4: Graphs

Assignment path

/data/assignments/dsa/assignment-4

Test cases

12 test cases · weighted

[Start grading ↗](#)

[View config](#)

Run status

batch 7f2a3c

23 / 45 jobs completed



Grading is asynchronous — this page polls status, actual execution happens via the RabbitMQ worker pool



04 · GET /api/assignments/{id}/results

Results — Assignment 3

DSA · Section 3

↺ Rejudge

⬇ Download .xlsx

Graded

42/45

Accepted

31

Flagged

2

Avg score

78%

Student	Score	Verdict	Graded
21i-0512	10/10	ACCEPTED	14:02
21i-0498	6/10	TIME_LIMIT_EXCEEDED	14:02
21i-0533	0/10	MALICIOUS_CODE	14:03
21i-0447	0/10	MEMORY_LIMIT_EXCEEDED	14:03
21i-0561	4/10	RUNTIME_ERROR	14:04
21i-0502	—	GRADING...	—



05 · [POST /api/assignments/{id}/rejudge](#) · [future](#)

Rejudge — Assignment 3

By student

Student ID

Re-runs all test cases for this student only. Overwrites prior result — `graded_at` is updated, no history kept.

[Rejudge student ↗](#)

By test case

Test case

Re-runs this single test case across all 45 students. Useful after fixing a bad expected-output file.

[Rejudge test case ↗](#)

Composite key (student_id, assignment_id, test_case_name) makes both operations directly addressable

06 · [/api/assignments/{id}/testcases](#) · [future](#)

Test cases — Assignment 3

[+ Add test case ↗](#)

Name	Input file	Expected output	Weight	
testcase_01	tc01.in	tc01.out	1.0	⋮
testcase_02	tc02.in	tc02.out	1.0	⋮
testcase_07 · edge case	tc07.in	tc07.out	2.5	⋮
testcase_12 · edge case	tc12.in	tc12.out	2.5	⋮

Weight configuration unified with config.json — resolved before this page was built, per the data-model decision

07 · GET/PUT /api/assignments/{id}/config · future

Config — Assignment 3

Grading

Language

cpp

Time limit (seconds)

2

Memory limit (MB)

256

Process limit

30

Sandbox

Network mode

none

CPU limit (cores)

0.5

pids-limit

100

Plagiarism check

Enabled · JPlag (cpp)

[Save changes ↗](#)

[Reset to defaults](#)

Form fields, not raw JSON editing — backed by a proper JSON parser (Jackson), not the original hand-rolled reader



08 · /api/assignments/{id}/plagiarism-results

Plagiarism — Assignment 3

[Run check](#)[Download report](#)

Pairs compared

990

Flagged (>70%)

3

Engine

JPlag · cpp

Submission A	Submission B	Similarity	
21i-0533	21i-0561	91%	
21i-0498	21i-0502	84%	
21i-0447	21i-0512	73%	

Pairwise data — deliberately not reusing the per-student results table layout